

Tax Multipliers in a DSGE Model*

Jason DeBacker[†]

Aerfate Haimiti[‡]

July 8, 2026

Abstract

This paper analyzes the stabilization properties of tax cuts in a DSGE model. We do this by simulating the a general equilibrium model with a rich set of tax levers in order to quantify tax multipliers over the short and medium-term. The novelty in this paper is that we focus on tax cuts that directly target consumption and investment, which are the most effective counter-cyclical fiscal policies. We find that the short-term multiplier associated with a cut to the consumption tax is larger than the multiplier associated with a cut to the labor income tax, and that the short-term multipliers associated with cuts to investment taxes are even larger. We compare these to the multipliers associated with cuts to broad taxes on capital and labor income, which are the focus of much of the existing literature. We also find that the medium-term multipliers associated with cuts to consumption and investment taxes are larger than the medium-term multipliers associated with cuts to broad taxes on capital and labor income. These results suggest that tax cuts that directly target consumption and investment are more effective counter-cyclical fiscal policies than tax cuts that target broad taxes on capital and labor income.

We study tax multipliers associated with taxes that directly stimulate private consumption spending or investment such as consumption taxes and investment tax credits. Consumption taxes have long been favored by tax economists for their relative efficiency vis-a-vis taxes on capital and labor. This paper attempts to argue that consumption taxes should also be favored by macroeconomists for their stabilization properties. Cuts to consumption taxes can quickly and effectively stimulate aggregate demand. The following analysis studies the quantitative impact of tax cuts in a dynamic stochastic general equilibrium (DSGE) model. In addition to consumption taxes, the model nests other tax policies, such as investment tax credits and changes to expensing and depreciation policies. These policies target investment by lowering the after-tax cost of capital. Historically, such instruments have been used by the U.S. to provide counter-cyclical fiscal policy (e.g. the Economic Recovery Act of 1981 increased the ability of businesses to take advantage of the investment tax credit). Despite their use, such policies have not been studied in a DSGE environment.

The goal of this paper is to study specific tax policies in a DSGE environment. A number of researchers have used DSGE models to study the government spending multiplier (e.g., Christiano, Eichenbaum and Rebelo (2010)), but less work has been done on tax multipliers. Those who provide insights on tax multipliers (e.g., Zubairy (2010), Kumhof, Coenen, Muir, Freedman, Mursula, Erceg, Furceri, Lalonde, Linde, Mourougane, Roberts,

*Preliminary– Please do not cite without author’s permission.

[†]University of South Carolina, Email: jason.debacker@moore.sc.edu

[‡]University of South Carolina, Email: haerfate@email.sc.edu

Laxton, de Resende, Roeger, Snudden, Trabandt and in 't Veld (2010)) focus on multipliers associated with broad taxes on capital and labor income. Such tax cuts are generally not the most effective counter-cyclical policies. We are the first to provide quantitative insights into the size of the multipliers associated with tax cuts that directly target consumption and investment.

Some related work to mention:

1. Kumhof et al. (2010) shows impulse response functions to a consumption tax cut. This paper seems to find that consumption taxes do well compared to other taxes and transfers, but don't do as well as gov't consumption and gov't investment
2. Leeper, Plante and Traum (2010) have consumption taxes, but don't allow them to vary with gov't debt
3. Zubairy (2010) doesn't not have consumption taxes, only capital and labor income taxes. Says few even have distortionary taxes in models of multipliers.
4. Traum and Yang (2010) allow for consumption taxes and allow them to vary. Not answering the same question as I want, but good to look at what they say about consumption taxes. In particular, how with cons taxes the consumer and producer price index will vary and what crowding out happens with consumption tax changes.
5. Coenen and Straub (2005) have consumption taxes, but don't use them as a policy instrument. They don't say anything interesting about consumption taxes.
6. Christiano et al. (2010) don't have consumption taxes and only focus on spending multipliers. I don't even think they have taxes. Fiscal and monetary policy rules look simple though- might want to take these.

The remainder of the paper is organized as follows. Section 1 describes the model. Section 1.7 describes the calibration and estimation strategy we use to identify parameter values. Section 2 presents the results. Section 8 concludes.

1 Model

1.1 Households

The economy is populated by a continuum of identical households. Each household maximizes expected lifetime utility over its habit-adjusted private consumption surplus $x_{c,t}$, hours worked h_t , and habit-adjusted government consumption surplus $x_{g,t}$:

$$E_0 \sum_{t=0}^{\infty} \beta^t d_t \left\{ \frac{[x_{c,t}^a (1 - h_t)^{1-a}]^{1-\gamma}}{1 - \gamma} + V(x_{g,t}) \right\} \quad (1.1)$$

where $\beta \in (0, 1)$ is the discount factor, d_t is an intertemporal preference shock, γ is the coefficient of relative risk aversion, and $a \in (0, 1)$ is the consumption expenditure share calibrated at the steady state. Utility over government consumption $V(x_{g,t})$ is separable from private utility, so government spending does not directly affect the marginal utility of private consumption or leisure. Following Zubairy (2010), the model incorporates deep habit formation in both private and government consumption as in Ravn, Schmitt-Grohé

and Uribe (2006), which generates counter-cyclical markups and allows private consumption to rise in response to a government spending shock.

The habit-adjusted consumption surpluses are:

$$x_{c,t} = c_t - b_c s_{C,t-1} \quad (1.2)$$

$$x_{g,t} = g_t - b_g s_{G,t-1} \quad (1.3)$$

where $b_c \in [0, 1)$ and $b_g \in [0, 1)$ govern the degree of deep habit formation in private and government consumption. The habit stocks evolve according to:

$$s_{C,t} = \theta^c s_{C,t-1} + (1 - \theta^c) c_t \quad (1.4)$$

$$s_{G,t} = \theta^g s_{G,t-1} + (1 - \theta^g) g_t \quad (1.5)$$

where $\theta^c, \theta^g \in [0, 1)$ govern the speed of habit stock adjustment.

Households supply labor, accumulate physical capital, hold government bonds, and receive dividend income from firms. The period-by-period real budget constraint is:

$$\begin{aligned} (1 + \tau_t^c) c_t + b_t + i_t = & \frac{[1 + (R_{t-1} - 1)(1 - \tau_t^i)] b_{t-1}}{\pi_t} + (1 - \tau_t^w) w_t h_t \\ & + (1 - \tau_t^k) r_t^k u_t k_{t-1} + \tau_t^k \delta^\tau k_{t-1}^\tau + \tau_t^k e_t^\tau i_t + \tau_t^{ic} i_t + (1 - \tau_t^d) \text{div}_t + x_t \end{aligned} \quad (1.6)$$

where b_t denotes real bond holdings, R_t is the gross nominal interest rate, $\pi_t \equiv P_t/P_{t-1}$ is gross price inflation, w_t is the real wage, r_t^k is the real rental rate of capital, u_t is the capital utilization rate, div_t are real firm dividends, and x_t is a lump-sum government transfer. The tax instruments are: τ_t^c (consumption tax), τ_t^w (labor income tax), τ_t^k (capital income tax), τ_t^i (interest income tax), τ_t^d (dividend income tax), and τ_t^{ic} (investment tax credit). Only the interest component $(R_{t-1} - 1)$ of the gross bond return is subject to the interest income tax. Capital income is measured net of tax depreciation deductions $\tau_t^k \delta^\tau k_{t-1}^\tau$, where δ^τ is the tax depreciation rate and k_t^τ is the tax basis of the capital stock, and net of immediate expensing $\tau_t^k e_t^\tau i_t$, where e_t^τ is the expensing rate.

Physical capital accumulates according to:

$$k_t = (1 - \delta) k_{t-1} + i_t \left[1 - S \left(\frac{\mu_t i_t}{i_{t-1}} \right) \right] \quad (1.7)$$

where δ is the constant economic depreciation rate and μ_t is a stochastic investment efficiency shock. Following Smets and Wouters (2003) and Zubairy (2010), the investment adjustment cost function takes the form $S \left(\frac{\mu_t i_t}{i_{t-1}} \right) = \frac{\kappa}{2} \left(\frac{\mu_t i_t}{i_{t-1}} - 1 \right)^2$, so that in the deterministic steady state $S = S' = 0$ and $S'' = \kappa > 0$.

The tax basis of the capital stock evolves as:

$$k_t^\tau = (1 - \delta^\tau) k_{t-1}^\tau + i_t (1 - e_t^\tau) \quad (1.8)$$

Only the non-expensed fraction $(1 - e_t^\tau)$ of new investment is added to the tax basis; the immediately expensed portion is deducted in the period of investment.

Household optimality conditions

The first-order condition with respect to consumption defines the marginal utility of income λ_t :

$$\lambda_t = \frac{d_t \cdot a \cdot [x_{c,t}^a (1 - h_t)^{1-a}]^{1-\gamma}}{x_{c,t} (1 + \tau_t^c)} \quad (1.9)$$

The consumption tax τ_t^c drives a wedge between the marginal utility of income and the marginal utility of the consumption surplus: a higher τ_t^c lowers λ_t for any given level of $x_{c,t}$ and h_t .

The intratemporal labor-leisure condition equates the marginal disutility of work to the after-markup real wage:

$$d_t (1 - a) \frac{[x_{c,t}^a (1 - h_t)^{1-a}]^{1-\gamma}}{1 - h_t} = \frac{\lambda_t w_t}{\tilde{w}_t} \quad (1.10)$$

where $\tilde{w}_t$ is the wage markup arising from monopolistic competition in the labor market. The labor income tax τ_t^w does not appear explicitly here because it is absorbed into the endogenous markup: the wage Phillips curve (1.28) implies $\tilde{w}_t = \eta_w / [(\eta_w - 1)(1 - \tau_t^w)]$ in the frictionless limit, so the effective wage $w_t / \tilde{w}_t$ already reflects the after-tax, after-markup return to labor.

The optimal utilization rate equates the after-tax marginal product of capital to the marginal cost of utilization:

$$(1 - \tau_t^k) r_t^k = \gamma_1 u_t^{\sigma_u} \quad (1.11)$$

where $\gamma_1 = (1 - \tau_k^{ss}) r_k^{ss}$ is pinned down at the steady state and σ_u governs the convexity of the utilization cost function $a(u_t) = \frac{\gamma_1}{1 + \sigma_u} (u_t^{1 + \sigma_u} - 1)$. Relative to Zubairy (2010), the term $\delta q_t \tau_t^k$ is absent from this condition because depreciation deductions are now tracked through the shadow value of the tax basis $v_{\tau,t}$ defined below, avoiding double-counting.

The capital Euler equation governs the optimal capital stock:

$$\lambda_t q_t = \beta E_t \lambda_{t+1} \left[(1 - \tau_{t+1}^k) r_{t+1}^k u_{t+1} - \frac{\gamma_1}{1 + \sigma_u} (u_{t+1}^{1 + \sigma_u} - 1) + q_{t+1} (1 - \delta) \right] \quad (1.12)$$

where q_t is Tobin's marginal q . Similarly, the term $\delta q_{t+1} \tau_{t+1}^k u_{t+1}$ present in Zubairy (2010) is omitted here for the same reason.

The investment Euler equation governs the optimal investment decision:

$$\lambda_t \left(1 - \tau_t^{ic} - \tau_t^k e_t^\tau \right) = \lambda_t q_t \left[1 - S \left(\frac{\mu_t i_t}{i_{t-1}} \right) - \frac{\mu_t i_t}{i_{t-1}} S' \left(\frac{\mu_t i_t}{i_{t-1}} \right) \right] + \beta E_t \lambda_{t+1} q_{t+1} \left(\frac{\mu_{t+1} i_{t+1}}{i_t} \right)^2 S' \left(\frac{\mu_{t+1} i_{t+1}}{i_t} \right) + \lambda_t v_{\tau,t} (1 - e_t^\tau) \quad (1.13)$$

The left-hand side is the net cost of investing one unit: the investment tax credit τ_t^{ic} and the immediate expensing saving $\tau_t^k e_t^\tau$ each reduce the effective outlay. The term $\lambda_t v_{\tau,t} (1 - e_t^\tau)$ on the right-hand side is the present value of future tax depreciation deductions on the non-expensed share of investment. The shadow value of the tax basis $v_{\tau,t}$ satisfies its own Euler equation:

$$\lambda_t v_{\tau,t} = \beta E_t \lambda_{t+1} \left[(1 - \delta^\tau) v_{\tau,t+1} + \tau_{t+1}^k \delta^\tau \right] \quad (1.14)$$

An additional unit of tax basis yields a deduction $\tau_{t+1}^k \delta^\tau$ next period and a depreciated stock $(1 - \delta^\tau)v_{\tau,t+1}$ of future deductions.

Finally, the bond Euler equation is:

$$\lambda_t = \beta E_t [1 + (R_t - 1)(1 - \tau_{t+1}^i)] \frac{\lambda_{t+1}}{\pi_{t+1}} \quad (1.15)$$

Only the interest earned $(R_t - 1)$ on bond holdings is subject to the interest income tax τ_{t+1}^i , so the relevant return is the after-tax net interest. This differs from Zubairy (2010), in which the pre-tax gross return R_t governs the bond Euler.

1.2 Firms

1.2.1 Final Goods Producers

A continuum of monopolistically competitive firms indexed by $i \in [0, 1]$ each produce a differentiated variety of output demanded by households, the government, and for investment. Because households and the government form deep habits over individual varieties (Section 1), the demand for variety i in the private consumption and government markets is:

$$c_{i,t} = \left(\frac{P_{i,t}}{P_t} \right)^{-\eta} x_{c,t} + b_c s_{C,i,t-1} \quad (1.16)$$

$$g_{i,t} = \left(\frac{P_{i,t}}{P_t} \right)^{-\eta} x_{g,t} + b_g s_{G,i,t-1} \quad (1.17)$$

Investment demand for variety i is $i_{i,t} = (P_{i,t}/P_t)^{-\eta}(y_t - c_t - g_t)$; there are no deep habits in investment. Here $\eta > 1$ is the elasticity of substitution across varieties and P_t is the aggregate price index. The habit-inelastic components $b_c s_{C,i,t-1}$ and $b_g s_{G,i,t-1}$ make a portion of demand perfectly price-inelastic, generating counter-cyclical markups: when aggregate demand expands, the price-elastic share rises, increasing effective demand elasticity and compressing markups. Firms therefore invest in their customer base by holding prices low today to earn higher profits on a larger habit stock in the future.

1.2.2 Intermediate Goods Producers

Each firm i produces output using capital services $u_t k_{i,t-1}$ and labor $h_{i,t}$ according to a Cobb-Douglas technology with a fixed operating cost ψ :

$$\tilde{y}_{i,t} = z_t (u_t k_{i,t-1})^\theta h_{i,t}^{1-\theta} - \psi \quad (1.18)$$

where z_t is an aggregate technology shock, θ is the capital income share, and ψ is calibrated to match a realistic profit-to-output ratio in steady state. The technology shock follows an AR(1) process:

$$\log z_t = \rho_z \log z_{t-1} + \varepsilon_t^z, \quad \varepsilon_t^z \sim \mathcal{N}(0, \sigma_z^2) \quad (1.19)$$

where $\rho_z \in [0, 1)$. Cost minimization at given factor prices yields the symmetric-equilibrium factor demand conditions:

$$w_t = mc_t \cdot z_t (1 - \theta) (u_t k_{t-1})^\theta h_t^{-\theta} \quad (1.20)$$

$$r_t^k = mc_t \cdot z_t \theta (u_t k_{t-1})^{\theta-1} h_t^{1-\theta} \quad (1.21)$$

where mc_t is the common real marginal cost across firms.

Following Rotemberg (1982), each firm faces a quadratic cost of changing its nominal price, which reduces output available for final demand. In symmetric equilibrium, aggregate output net of price adjustment costs is:

$$y_t = z_t(u_t k_{t-1})^\theta h_t^{1-\theta} - \psi - \frac{\alpha_P}{2}(\pi_t - \bar{\pi})^2 \quad (1.22)$$

where $\alpha_P > 0$ governs the magnitude of price adjustment costs and $\bar{\pi}$ is the steady-state gross inflation rate.

Because deep habits create dynamic pricing incentives, each firm tracks the shadow value of its customer base in each demand segment. In symmetric equilibrium, the shadow values $\tilde{v}_t^c$, $\tilde{v}_t^g$, and $\tilde{v}_t^i$ for the private consumption, government consumption, and investment markets satisfy:

$$1 - mc_t - \tilde{v}_t^i = 0 \quad (1.23)$$

$$\frac{1 - mc_t - \tilde{v}_t^c}{\theta^c - 1} = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[b_c \tilde{v}_{t+1}^c + \frac{\theta^c}{\theta^c - 1} (1 - mc_{t+1} - \tilde{v}_{t+1}^c) \right] \quad (1.24)$$

$$\frac{1 - mc_t - \tilde{v}_t^g}{\theta^g - 1} = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[b_g \tilde{v}_{t+1}^g + \frac{\theta^g}{\theta^g - 1} (1 - mc_{t+1} - \tilde{v}_{t+1}^g) \right] \quad (1.25)$$

Equation (1.23) states that the investment market, lacking habit-inelastic demand, yields a zero effective markup. Equations (1.24)–(1.25) are forward-looking Euler equations: a firm accepts a lower current markup to build its habit stock, earning higher future rents.

These conditions together with the demand structure yield the aggregate price Phillips curve:

$$\eta(\tilde{v}_t^c x_{c,t} + \tilde{v}_t^g x_{g,t} + \tilde{v}_t^i (y_t - c_t - g_t)) + \alpha_P \pi_t (\pi_t - \bar{\pi}) - y_t = \alpha_P \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1} (\pi_{t+1} - \bar{\pi}) \quad (1.26)$$

Each household is also a monopolistic supplier of a differentiated labor variety and faces quadratic wage adjustment costs. Defining gross wage inflation $\pi_t^w \equiv (w_t/w_{t-1})\pi_t$, the wage Phillips curve is:

$$\pi_t^w = \frac{w_t}{w_{t-1}} \pi_t \quad (1.27)$$

$$(\eta_w - 1)(1 - \tau_t^w)h_t + \alpha_W \pi_t^w (\pi_t^w - \bar{\pi}) - \frac{\eta_w h_t}{\tilde{w}_t} = \alpha_W \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1}^w (\pi_{t+1}^w - \bar{\pi}) \quad (1.28)$$

where $\eta_w > 1$ is the elasticity of substitution across labor varieties and α_W governs wage adjustment costs. The labor income tax τ_t^w enters the wage Phillips curve because it affects the effective value of the marginal product of labor to the household.

Firm dividends are the residual of net output after paying all factors:

$$\text{div}_t = y_t - w_t h_t - r_t^k u_t k_{t-1} \quad (1.29)$$

The aggregate resource constraint equates gross production (net of fixed costs) to all expenditure uses plus utilization and price and wage adjustment costs:

$$z_t(u_t k_{t-1})^\theta h_t^{1-\theta} - \psi = c_t + g_t + i_t + \frac{\gamma_1}{1 + \sigma_u} (u_t^{1+\sigma_u} - 1) k_{t-1} + \frac{\alpha_P}{2} (\pi_t - \bar{\pi})^2 + \frac{\alpha_W}{2} (\pi_t^w - \bar{\pi})^2 w_t \quad (1.30)$$

1.3 Government/Fiscal Authority

The government issues one-period nominal bonds, purchases goods, pays lump-sum transfers to households, and collects tax revenue from all six tax instruments. The real government budget constraint is:

$$b_t = \frac{R_{t-1} b_{t-1}}{\pi_t} + g_t + x_t - taxrev_t + \varepsilon_t^b \quad (1.31)$$

where b_t is real government debt, g_t is real government purchases, x_t is real lump-sum transfers, $taxrev_t$ is total real tax revenue, and ε_t^b is an i.i.d. residual added to the budget identity. Tax revenue is:

$$\begin{aligned} taxrev_t = & \tau_t^c c_t + \tau_t^w w_t h_t + \tau_t^k \left(r_t^k u_t k_{t-1} - \delta^\tau k_{t-1}^\tau - e_t^\tau i_t \right) \\ & + \tau_t^i (R_{t-1} - 1) \frac{b_{t-1}}{\pi_t} + \tau_t^d \text{div}_t - \tau_t^{ic} i_t \end{aligned} \quad (1.32)$$

The six components are: consumption tax revenue ($\tau_t^c c_t$), labor income tax revenue ($\tau_t^w w_t h_t$), capital income tax revenue net of tax depreciation deductions and immediate expensing ($\tau_t^k(\cdot)$), interest income tax revenue on bond interest ($\tau_t^i (R_{t-1} - 1) b_{t-1} / \pi_t$), dividend tax revenue ($\tau_t^d \text{div}_t$), and investment tax credit outlays ($-\tau_t^{ic} i_t$).

Fiscal rules

Each fiscal instrument follows an autoregressive rule that allows feedback from lagged government debt and lagged output, following Zubairy (2010). Debt feedback ensures intertemporal fiscal solvency; output feedback captures automatic stabilizers. Because g_t and x_t are expenditures that the government cuts when debt is elevated, their debt-feedback coefficients enter with a negative sign. Tax rates carry a positive debt-feedback sign, rising to close the fiscal gap.

Government purchases and lump-sum transfers follow:

$$\log\left(\frac{g_t}{\bar{g}}\right) = \rho_g \log\left(\frac{g_{t-1}}{\bar{g}}\right) - \rho_{g,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{g,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^g \quad (1.33)$$

$$\log\left(\frac{x_t}{\bar{x}}\right) = \rho_x \log\left(\frac{x_{t-1}}{\bar{x}}\right) - \rho_{x,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{x,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^x \quad (1.34)$$

The rules for the capital and labor income tax rates, which follow Zubairy (2010), are:

$$\log\left(\frac{\tau_t^k}{\bar{\tau}^k}\right) = \rho_k \log\left(\frac{\tau_{t-1}^k}{\bar{\tau}^k}\right) + \rho_{k,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{k,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^k \quad (1.35)$$

$$\log\left(\frac{\tau_t^w}{\bar{\tau}^w}\right) = \rho_w \log\left(\frac{\tau_{t-1}^w}{\bar{\tau}^w}\right) + \rho_{w,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{w,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \phi \varepsilon_t^k + \varepsilon_t^w \quad (1.36)$$

The term $\phi \varepsilon_t^k$ in the labor tax rule allows capital tax shocks to generate correlated movements in the labor tax rate, capturing the empirical co-movement between capital and labor income tax changes.

The four extension fiscal rules for the additional tax instruments take the same log-ratio form:

$$\log\left(\frac{\tau_t^c}{\bar{\tau}^c}\right) = \rho_c \log\left(\frac{\tau_{t-1}^c}{\bar{\tau}^c}\right) + \rho_{c,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{c,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^c \quad (1.37)$$

$$\log\left(\frac{\tau_t^i}{\bar{\tau}^i}\right) = \rho_i \log\left(\frac{\tau_{t-1}^i}{\bar{\tau}^i}\right) + \rho_{i,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{i,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^i \quad (1.38)$$

$$\log\left(\frac{\tau_t^d}{\bar{\tau}^d}\right) = \rho_d \log\left(\frac{\tau_{t-1}^d}{\bar{\tau}^d}\right) + \rho_{d,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{d,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \phi_{wd} \varepsilon_t^w + \varepsilon_t^d \quad (1.39)$$

The dividend tax rule includes $\phi_{wd} \varepsilon_t^w$, capturing the empirical co-movement between changes in labor and dividend income tax rates. The rules for the investment tax credit and the expensing rate are specified in level-deviation form because the log-ratio form is numerically unstable when the underlying rate can take values near zero:

$$\tau_t^{ic} - \bar{\tau}^{ic} = \rho_{ic}(\tau_{t-1}^{ic} - \bar{\tau}^{ic}) + \rho_{ic,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{ic,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^{ic} \quad (1.40)$$

$$e_t^\tau - \bar{e}^\tau = \rho_e(e_{t-1}^\tau - \bar{e}^\tau) + \rho_{e,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{e,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^e \quad (1.41)$$

1.4 Monetary Authority

The monetary authority sets the gross nominal interest rate R_t according to a Taylor rule with interest rate smoothing, following Zubairy (2010):

$$\log\left(\frac{R_t}{\bar{R}}\right) = \alpha_R \log\left(\frac{R_{t-1}}{\bar{R}}\right) + (1 - \alpha_R) \left[\alpha_\pi \log\left(\frac{\pi_t}{\bar{\pi}}\right) + \alpha_Y \log\left(\frac{y_t}{\bar{y}}\right) \right] + \varepsilon_t^m \quad (1.42)$$

where $\alpha_R \in (0, 1)$ governs interest rate inertia, $\alpha_\pi > 1$ is the response to inflation deviations from steady state (satisfying the Taylor principle), $\alpha_Y > 0$ is the response to output deviations, and ε_t^m is a monetary policy shock. The interest income tax τ_t^i does not enter the Taylor rule directly; it affects the economy through the household bond Euler equation (1.15), which links the pre-tax policy rate R_t set by the central bank to the after-tax return relevant for household saving decisions.

1.5 Equilibrium

Definition (Competitive Rational Expectations Equilibrium). Given exogenous shock processes $\{z_t, d_t, \mu_t\}_{t=0}^\infty$ and initial conditions $(k_{-1}, k_{-1}^\tau, s_{C,-1}, s_{G,-1}, b_{-1})$, a *competitive rational expectations equilibrium* is a collection of sequences of quantities

$$\{c_t, i_t, g_t, x_t, y_t, h_t, u_t, k_t, k_t^\tau, b_t, \text{div}_t, s_{C,t}, s_{G,t}, x_{c,t}, x_{g,t}\}_{t=0}^\infty,$$

prices and shadow values

$$\{w_t, r_t^k, R_t, \pi_t, \pi_t^w, mc_t, q_t, \lambda_t, v_{\tau,t}, \tilde{w}_t, \tilde{v}_t^c, \tilde{v}_t^g, \tilde{v}_t^i\}_{t=0}^\infty,$$

and fiscal instrument sequences $\{\tau_t^c, \tau_t^w, \tau_t^k, \tau_t^i, \tau_t^d, \tau_t^{ic}, e_t^\tau\}_{t=0}^\infty$ such that:

- (i) Taking prices and tax rates as given, the representative household maximizes (1.1) subject to the budget constraint (1.6), capital accumulation (1.7), and tax basis accumulation (1.8), yielding the first-order conditions (1.9)–(1.15).

- (ii) Taking wages w_t and rental rates r_t^k as given, each intermediate goods firm $i \in [0, 1]$ minimizes costs, yielding factor demands (1.20)–(1.21), and sets its nominal price to maximize the present discounted value of dividends subject to Rotemberg adjustment costs and the deep habits demand structure (1.16)–(1.17), yielding customer-base shadow values (1.23)–(1.25) and the price Phillips curve (1.26).
- (iii) Each household sets its nominal wage to maximize utility subject to Rotemberg wage adjustment costs, yielding the wage Phillips curve (1.28).
- (iv) The government satisfies its budget constraint (1.31) with tax revenue defined by (1.32), and each fiscal instrument follows its policy rule (1.33)–(1.41).
- (v) The central bank sets the gross nominal interest rate R_t according to the Taylor rule (1.42).
- (vi) Labor, capital, bond, and goods markets clear according to conditions (1.43)–(1.45).
- (vii) Expectations $E_t[\cdot]$ are rational: agents' beliefs are consistent with the probability distribution implied by the model.

1.5.1 Market Clearing

In a symmetric equilibrium all intermediate goods firms make identical factor choices, so labor and capital markets clear when household factor supply equals aggregate firm demand:

$$h_t = \int_0^1 h_{i,t} di \quad (1.43)$$

$$k_{t-1} = \int_0^1 k_{i,t-1} di \quad (1.44)$$

In symmetric equilibrium these reduce to $h_{i,t} = h_t$ and $k_{i,t-1} = k_{t-1}$ for all $i \in [0, 1]$.

The bond market clears when household bond holdings equal the government's outstanding debt. Since both the household budget constraint (1.6) and the government budget constraint (1.31) use the same variable b_t , this condition holds by construction.

Goods market clearing requires total net output to equal total absorption. Combining the output definition (1.22) with the resource constraint (1.30) yields:

$$y_t = c_t + g_t + i_t + \frac{\gamma_1}{1 + \sigma_u} (u_t^{1+\sigma_u} - 1) k_{t-1} + \frac{\alpha_W}{2} (\pi_t^w - \bar{\pi})^2 w_t \quad (1.45)$$

Output y_t is already net of the Rotemberg price adjustment cost (see equation (1.22)), so only the utilization cost and wage adjustment cost appear on the right-hand side. By Walras' law, the bond market clearing condition is redundant given (1.43), (1.44), and (1.45).

1.6 Solving the Model

The equilibrium conditions described in Section 1 form a system of 46 nonlinear expectational difference equations in 46 structural endogenous variables, driven by 13 exogenous shock processes. The model is solved and estimated in Dynare using a three-step procedure: analytical computation of the deterministic steady state, first-order perturbation approximation of the equilibrium system, and Bayesian estimation via the Kalman filter and Metropolis-Hastings algorithm.

1.6.1 Steady State

The model's deterministic steady state is computed analytically by setting all shocks to their unconditional means ($z = d = \mu = 1$; all fiscal innovations zero) and imposing stationarity. The calibrated parameters in Table 1 jointly pin down the steady-state tax rates and the aggregate ratios $\bar{g}/\bar{y}$ and $\bar{b}/\bar{y}$, from which all remaining steady-state values follow in closed form. In particular, the steady-state rental rate $\bar{r}^k$, wage $\bar{w}$, output $\bar{y}$, consumption $\bar{c}$, investment $\bar{i}$, and government debt $\bar{b}$ are each expressed as analytic functions of the calibrated parameters. The fixed operating cost ψ is then set residually to match the target dividend share $\bar{\text{div}}/\bar{y} = 0.05$. Because several estimated parameters (deep habit parameters, risk aversion, Rotemberg costs) enter the steady-state expressions, all steady-state values are recomputed at each draw of the Metropolis-Hastings algorithm.

1.6.2 First-Order Perturbation Approximation

Let y_t denote the n_y -vector of endogenous variables and $u_t \sim \mathcal{N}(0, \Sigma)$ the n_u -vector of structural shocks. The equilibrium conditions can be written compactly as:

$$E_t[f(y_{t+1}, y_t, y_{t-1}, u_t; \Theta)] = 0 \quad (1.46)$$

where Θ is the structural parameter vector. Dynare takes a first-order Taylor expansion of (1.46) around the deterministic steady state $(\bar{y}, \mathbf{0})$, yielding the linear policy function:

$$\hat{y}_t = A(\Theta) \hat{y}_{t-1} + B(\Theta) u_t \quad (1.47)$$

where $\hat{y}_t \equiv y_t - \bar{y}$ denotes deviations from steady state and the coefficient matrices A and B are functions of the structural parameters. Dynare computes A and B using the generalized Schur (QZ) decomposition (Klein 2000), which handles both predetermined and forward-looking variables. A unique stationary rational expectations solution exists if and only if the number of eigenvalues of A outside the unit circle equals the number of non-predetermined (forward-looking) variables — the Blanchard-Kahn conditions (Blanchard and Kahn 1980). Dynare verifies these conditions numerically at the posterior mode and at every point during estimation at which the likelihood is evaluated.

By Walras' law, one market clearing condition is redundant; Dynare drops the goods market clearing equation (1.45) from the system prior to solving, leaving 45 independent structural equations. The policy function (1.47) therefore maps the 13 endogenous state variables ($k_t, k_t^\tau, b_t, s_{C,t}, s_{G,t}, \tau_t^k, \tau_t^w, \tau_t^c, \tau_t^{ic}, \tau_t^i, \tau_t^d, e_t^\tau, w_t$) and 13 exogenous shock states into the full set of endogenous variables.

1.6.3 Likelihood Evaluation and Bayesian Estimation

The policy function (1.47) together with the 12 measurement equations (Section 1.7) form a linear Gaussian state-space system. The Kalman filter recursively evaluates the prediction-error decomposition of the log-likelihood:

$$\log \mathcal{L}(Z^T \mid \Theta) = -\frac{T \cdot n_{obs}}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T [\log |F_t| + \nu_t' F_t^{-1} \nu_t] \quad (1.48)$$

where $\nu_t = z_t^{obs} - E_{t-1}[z_t^{obs}]$ is the n_{obs} -vector of one-step-ahead prediction errors and F_t is its conditional variance. The Kalman filter is initialized with a diffuse prior on the state

vector. Observed data are demeaned prior to estimation to handle potential mismatches between model steady states and sample means.

Combining the likelihood with the prior distributions specified in Table 2 yields the log-posterior kernel:

$$\log p(\Theta \mid Z^T) = \log \mathcal{L}(Z^T \mid \Theta) + \log p(\Theta) + \text{const.} \quad (1.49)$$

The posterior mode is located via a Monte Carlo-based numerical optimization routine. The negative inverse Hessian at the mode provides an estimate of the posterior covariance, which is used to calibrate the scaling of a Gaussian random-walk proposal distribution for the Metropolis-Hastings algorithm. Two parallel chains of 1,500,000 draws each are generated, with the first one-third discarded as burn-in, yielding approximately 2,000,000 retained draws for posterior inference. Impulse response functions and fiscal multipliers are computed from draws of $A(\Theta)$ and $B(\Theta)$ over the retained posterior sample.

1.7 Calibration and Estimation

The model's parameter vector is partitioned into two groups. A subset of parameters is *calibrated*: fixed prior to estimation at values drawn from the literature or from long-run data targets, because they are weakly identified by the business-cycle observables used in estimation. The remaining parameters are *estimated* via Bayesian methods using U.S. quarterly data, following Zubairy (2010) and Smets and Wouters (2003). We use Dynare to compute the posterior mode via a Monte Carlo-based optimization routine and to characterize the posterior distribution via the Metropolis-Hastings algorithm with 1,500,000 draws across two chains, discarding the first one-third of draws as burn-in.

1.7.1 Calibrated Parameters

Table 1 reports the calibrated parameters. The discount factor β is set to imply a 3% annual real interest rate. The quarterly economic depreciation rate $\delta = 0.025$ corresponds to a standard 10% annual rate. The capital income share $\theta = 0.30$ matches the average labor share of income in U.S. national accounts. The price elasticity of substitution $\eta = 5.3$ implies a steady-state markup of roughly 23%, while the labor variety elasticity $\eta_w = 21.0$ is calibrated to match a 5% steady-state wage markup consistent with the Rotemberg wage-setting framework.

Steady-state hours worked $\bar{h} = 0.50$ normalizes the time endowment to unity and follows Zubairy (2010). The government spending share $\bar{g}/\bar{y} = 0.21$ and debt-to-output ratio $\bar{b}/\bar{y} = 0.32$ are set to the sample averages over 1961Q1–2013Q4. Steady-state inflation is normalized to $\bar{\pi} = 1$ (zero net inflation).

Steady-state tax rates are calibrated to the sample averages over the same period. The capital income tax $\bar{\tau}^k = 0.42$ and labor income tax $\bar{\tau}^w = 0.22$ follow the effective tax rate construction of Jones (2002). The consumption tax $\bar{\tau}^c = 0.06$ follows the construction of Mendoza, Razin and Tesar (1994) and reflects the effective tax rate on private consumption expenditures. The interest income and dividend tax rates $\bar{\tau}^i = 0.25$ and $\bar{\tau}^d = 0.30$ are the sample averages of the qualified interest and dividend marginal tax rates from the McGrattan TAXSIM series. The investment tax credit rate $\bar{\tau}^{ic} = 0.02$ and expensing rate $\bar{e}^\tau = 0.06$ are sample averages of the House and Shapiro (2008) ITC series and the BEA asset-weighted bonus depreciation series, respectively. The tax depreciation rate $\delta^\tau = 0.025$ equals the economic depreciation rate as a baseline. The dividend share of output $\bar{\text{div}}/\bar{y} =$

0.05 is set to the sample average; the fixed operating cost ψ in the production function is determined residually to hit this target in steady state.

1.7.2 Data

We use quarterly U.S. data from 1961Q1 to 2013Q4. Real per-capita consumption, investment, government purchases, and gross domestic product are constructed from BEA National Income and Product Accounts and the U.S. Census population series, then log-linear-detrended and expressed as percent deviations from trend. Real federal debt is the Dallas Fed market-value series (MVPHGFD027MNFRBDAL). Inflation is measured as gross GDP deflator inflation, and the interest rate is the Federal Funds Rate.

Effective tax rates on labor and capital income follow the NIPA-based construction of Jones (2002), and the consumption tax rate follows Mendoza et al. (1994). The dividend tax rate is taken from the McGrattan TAXSIM series (average marginal rate on qualified dividends), and the interest income tax rate is the corresponding TAXSIM series on interest income. The investment tax credit rate is constructed following House and Shapiro (2008) as a weighted aggregation of the statutory ITC schedule across equipment types. The expensing rate is the BEA asset-weighted average of statutory bonus depreciation provisions. Sample coverage is dictated by the availability of the TAXSIM series, which begins in 1960 and ends in 2013. All tax rate series are constructed as fractional deviations from their sample means.

Two features of the data motivate specific modeling choices. First, the 2008Q4 observation of market-value federal debt exhibits a large jump driven by valuation and issuance movements that are not modeled explicitly; the debt residual ε_t^b in the government budget constraint absorbs this so that the fiscal rules do not reconcile the jump through counterfactual movements in the tax instruments. Second, the ITC rate was inactive for most quarters after 1986 and bonus depreciation was inactive prior to 2001, so both τ_t^{ic} and e_t^τ take values at or near zero for extended periods of the sample. This is the reason the corresponding fiscal rules use the level-deviation form (equations 1.40 and 1.41) rather than the log-ratio form used for the other tax instruments.

1.7.3 Bayesian Estimation

The remaining 58 parameters are estimated jointly using Bayesian techniques. We match 12 quarterly observables from U.S. data to 12 of the model's measurement equations:

1. $\hat{c}_t = 100(c_t - \bar{c})/\bar{c}$ — private consumption (percent deviation from SS)
2. $\hat{i}_t = 100(i_t - \bar{i})/\bar{i}$ — private investment (percent deviation)
3. $\hat{g}_t = 100(g_t - \bar{g})/\bar{g}$ — government purchases (percent deviation)
4. $\pi_t - \bar{\pi}$ — gross price inflation (level deviation)
5. $R_t - \bar{R}$ — gross nominal interest rate (level deviation)
6. $(\tau_t^k - \bar{\tau}^k)/\bar{\tau}^k$ — capital income tax rate (fractional deviation)
7. $(\tau_t^w - \bar{\tau}^w)/\bar{\tau}^w$ — labor income tax rate (fractional deviation)
8. $100(b_t - \bar{b})/\bar{b}$ — government debt (percent deviation)

9. $(\tau_t^c - \bar{\tau}^c)/\bar{\tau}^c$ — consumption tax rate (fractional deviation)
10. $(\tau_t^d - \bar{\tau}^d)/\bar{\tau}^d$ — dividend tax rate (fractional deviation)
11. $\tau_t^{ic} - \bar{\tau}^{ic}$ — investment tax credit rate (level deviation)
12. $e_t^\tau - \bar{e}^\tau$ — expensing rate (level deviation)

Transfer payments x_t are excluded from the observable set because the 2008–2010 ARRA stimulus produced outliers of a magnitude the linearized model cannot fit; without this exclusion, the transfer shock variance inflates and distorts other fiscal-rule estimates. The interest income tax rate τ_t^i is also excluded because it has a 0.77 correlation with the dividend tax rate at the quarterly frequency, both being constructed from the same TAXSIM series; including both induces collinearity in the measurement equations. The shocks ε_t^{tr} and ε_t^i are retained in the model as unobserved structural shocks identified through cross-equation restrictions. Together with the debt-residual shock ε_t^b , the model has 14 structural shocks and 12 observables. Observed data are demeaned prior to estimation (`prefilter` = 1) to accommodate potential mismatches between model steady states and sample means.

Table 2 reports the estimated parameters along with their prior distributions. Prior choices closely follow Zubairy (2010) for the 28 parameters common to both models. Deep habit parameters b_c , θ^c , b_g , θ^g receive Beta priors on $(0, 1)$ consistent with their interpretation as persistence parameters. The risk aversion coefficient γ , utilization curvature σ_u , investment adjustment cost κ , and nominal rigidity parameters α_P , α_W receive Normal priors centered near values that generate moderate nominal rigidity. Fiscal rule AR coefficients ρ_j all receive Beta(0.7, 0.2) priors reflecting high but uncertain persistence. Debt-feedback coefficients for tax rules receive Gamma priors (positive), while those for spending rules receive Normal priors centered below zero (negative feedback). Output-feedback coefficients receive Normal priors centered near zero. All shock standard deviations receive inverse-Gamma priors. The 15 parameters for the five extension fiscal rules (ρ_c , ρ_{ic} , ρ_{ti} , ρ_{td} , $\rho_{e\tau}$ and their debt and output responses) and the cross-shock correlation ϕ_{wd} are new to this paper; their priors are set symmetrically with the analogous Zubairy parameters.

2 Results

2.1 Computing the Multipliers

For each fiscal instrument j , we compute the present-value multiplier at horizon H as

$$M_j(H) = \frac{\pm \sum_{t=0}^H \bar{R}^{-t} \Delta Y_t^{(j)}}{D_j(H)} \quad (2.1)$$

where $\Delta Y_t^{(j)}$ is the response of output at horizon t to a one-standard-deviation shock ε_t^j , $\bar{R}$ is the steady-state gross nominal interest rate, and the sign in the numerator is chosen so that the multiplier is positive for expansionary policy (a tax cut or spending increase). The denominator $D_j(H)$ is the change in the fiscal cost of the policy, for which we consider two conventions.

Under the first convention, which follows Zubairy (2010), the denominator is the present-value cumulative change in total tax revenue over horizon H :

$$D_j^{base}(H) = \sum_{t=0}^H \bar{R}^{-t} \Delta taxrev_t^{(j)} \quad (2.2)$$

Because $taxrev_t$ is total revenue net of ITC outlays (equation 1.32), this denominator captures the full fiscal impact of the policy including behavioral spillovers to bases of other tax instruments. We refer to this as the “total-revenue” or “base” normalization.

Under the second convention, the denominator is the year-1 (first four quarters) discounted cumulative change in the instrument’s own revenue:

$$D_j^{rev}(H) = \sum_{t=0}^3 \bar{R}^{-t} \Delta Rev_{j,t} \quad (2.3)$$

where $Rev_{j,t}$ is the component of $taxrev_t$ collected by instrument j (for the ITC, the fiscal outlay $\tau_t^{ic} i_t$). For the expensing rate, whose fiscal impact operates through the capital tax base, we use the year-1 change in capital income tax revenue as the denominator. We introduce this normalization because the seven tax instruments in our model differ substantially in the size of their bases: a one-percentage-point change in the consumption tax has a much smaller dollar revenue impact than a one-percentage-point change in the capital income tax. The total-revenue denominator absorbs this asymmetry through cross-instrument feedback in the fiscal rules, whereas the own-revenue denominator preserves it, delivering a per-dollar comparison closer in spirit to how such policies are scored by fiscal analysts.

We report both sets of multipliers in Tables 3 and 4. Point estimates are posterior medians and intervals in brackets are 5th and 95th percentiles of the posterior, computed from 1,000 draws sampled from the retained MCMC chains.

2.2 Short Run Multipliers

Table 3 reports the multipliers at horizons of 1, 4, 12, and 20 quarters under the total-revenue normalization. Panel A covers the three instruments most commonly studied in the fiscal multiplier literature; Panel B reports the additional tax instruments that are the focus of this paper; and Panel C reproduces the corresponding values from Zubairy (2010), Table 2, for comparison with a benchmark medium-scale DSGE estimate.

The government spending multiplier in Panel A is 1.09 on impact and declines to 0.35 by 20 quarters. This profile is broadly similar to the Zubairy estimates in Panel C, though our medium-run values decay somewhat faster. The labor tax multiplier is small on impact (0.22) and rises with horizon, peaking near 0.79 at 12 quarters, which seems consistent with the pattern that broad income tax cuts take time to work through labor supply and capital accumulation. The capital tax multiplier is large: 0.90 on impact and 3.08 at 12 quarters. This is notably larger than the corresponding Zubairy value (0.34 to 0.52) and may reflect the estimated persistence of the capital tax process combined with slow adjustment of the capital stock.

Panel B reports the multipliers for the tax instruments that are the focus of this paper. The investment tax credit produces the largest short-run response, at 0.87 on impact and 3.71 at 12 quarters. The expensing rate is smaller but positive throughout, ranging from 0.28 to 0.65. The consumption tax multiplier is modest on impact (0.28) and remains small at longer horizons, turning slightly negative by 20 quarters; a fuller discussion is offered in Section 2.4 below. The dividend tax multiplier is small and turns slightly negative at long horizons; because dividends are a residual object, the dividend tax operates only through the household’s after-tax income and does not directly enter any first-order condition. The transfer multiplier is essentially zero throughout, consistent with Ricardian equivalence in

the linearized model. The interest income tax multiplier is negative across all horizons, which is unusual; a partial explanation may be that this instrument is not directly observed in the estimation and is identified only through cross-equation restrictions. A pattern that seems to emerge across Panels A and B is that the largest multipliers are found among instruments that directly affect the marginal cost of investment: the capital income tax, the investment tax credit, and, to a lesser extent, the expensing rate.

Table 4 reports the same multipliers under the own-revenue normalization. The broad ranking is preserved: capital tax and ITC remain the largest, labor and consumption taxes remain smaller. The main change is that the levels of the investment-related multipliers move up further under this denominator, since these instruments involve substantial revenue changes concentrated in the first year, while the total-revenue denominator dilutes them through cross-instrument feedback.

Figures 1 through 7 plot the posterior impulse response functions of output, consumption, investment, hours, inflation, and the interest rate to a one-standard-deviation innovation in each fiscal instrument. Solid lines are posterior medians and shaded regions are 5–95 percentile bands. For the tax-rate instruments, a positive innovation raises the tax rate, so an output decline in the figure corresponds to a positive multiplier in the table. The government spending IRF (Figure 1) shows the expected pattern, with output rising on impact and consumption crowded in by the deep-habits mechanism. The capital tax IRF (Figure 3) shows a sharp and persistent decline in investment, driving a large output response that builds over several quarters as the capital stock adjusts. The investment tax credit IRF (Figure 5) is qualitatively similar but with a faster investment response, since the credit lowers the marginal cost of investment more directly.

2.3 Multipliers over the Medium Term

Figures 8 and 9 plot the present-value multipliers as a function of the horizon H up to 40 quarters after the shock, under the two normalizations. Focusing on the total-revenue version in Figure 8, the capital tax and investment tax credit continue to build past the 12-quarter mark before flattening, which seems consistent with the intuition that policies affecting the return on capital work through a slow-moving accumulation channel. Government spending has the largest impact multiplier but decays steadily over the horizon, which may reflect crowding-out through higher interest rates and lower private consumption. The labor and consumption tax multipliers remain more modest throughout. Under the own-revenue normalization in Figure 9, the profiles look qualitatively similar but with the investment-related multipliers scaled up further; the broad ranking is unchanged.

2.4 The Small Consumption Tax Multiplier

The consumption tax multiplier is small relative to what one might expect from the direct pass-through of a sales tax cut into consumer prices. Three features of the estimated model appear to contribute to this result, and none of them are easy to disentangle in the estimation.

Deep habits pin down consumption tightly to its own history. The posterior for b_c sits near 0.80 and θ^c near 0.75, so habit-adjusted consumption $x_t^c = c_t - b^c s_{t-1}^C$ is small relative to c_t , and the household resists moving current consumption away from the habit stock. When τ_t^c falls, the real price of consumption drops, but the household still smooths. Much of the tax cut seems to be absorbed into savings rather than into higher current spending. A

specification without deep habits, or with a hand-to-mouth share as in Coenen and Straub (2005), would likely deliver a larger short-run multiplier through this channel.

The persistence of the tax process itself is only moderate in the posterior. Our estimate of ρ_c is around 0.78, implying a fairly short half-life for a shock to the consumption tax rule. Present-value multipliers are sensitive to how long a cut is expected to remain in place; the near-permanent capital tax shocks in this estimation ($\rho_k \approx 0.94$) produce cumulative output responses that build over many quarters, whereas a shorter-lived consumption tax cut simply does not accumulate as much output effect.

Beyond these two mechanisms, the consumption tax has no direct investment channel. It does not enter the after-tax return on capital or the effective cost of investment goods. Its transmission runs only through the labor-leisure margin, via the real wage measured in consumption units, and through intertemporal substitution in consumption. Both of these channels are muted by the deep-habit mechanism already discussed.

The small multiplier is best read not as a puzzle but as a feature of a single-agent model with strong estimated consumption habits, moderate persistence of the tax rule, and no direct effect on investment. Getting a larger consumption tax multiplier out of the model would probably require either a shorter habit horizon, a more persistent tax rule, or a two-agent structure with a positive share of hand-to-mouth households – the last of these being the more common route in the recent literature.

3 Sensitivity to Government Financing

FIGURE: graph of multiplier over time given how gov't financing temporary tax cuts (with higher levels of debt going forward or with paying back by increasing taxes). Maybe 3D graph with multiplier, years from cut, and year to pay back debt as axes.

4 Sensitivity to Monetary Policy

Talk about multipliers at and away from the zero-bound. What happens if hold interest rates constant?

5 Dead-weight Loss and Fiscal Policy

1. TABLE: table of DWL for each of policy responses
2. FIGURE: graph of DWL over time for policy response

Do some revenue neutral changes in tax policy and calculate welfare....

6 Sensitivity to Parameters

How do results change with change to key parameters...

7 Discussion

Note that hard to know preferences for gov't spending, but imagine that people know better what want, cutting taxes may be politically easier than increasing spending. Note other weaknesses of the modeling assumptions.

If consumption taxes are useful stabilization tool, and not just a tool for more revenue, countries like the US that don't have them should consider implementing them. This is especially true if the zero bound is a concern, since consumption tax cuts can be effective even when interest rates are at the zero bound. Just like a non-zero inflation target can help give monetary policy more room to maneuver, a non-zero consumption tax can give fiscal policy more room to maneuver.

8 Conclusion

References

- Blanchard, Olivier Jean and Charles M Kahn**, “The solution of linear difference models under rational expectations,” *Econometrica: Journal of the Econometric Society*, 1980, pp. 1305–1311.
- Christiano, Lawrence, Martin Eichenbaum, and Sergio Rebelo**, “When is the government spending multiplier large?,” *NBER Working Paper*, December 2010, (#15394).
- Coenen, Gunter and Roland Straub**, “Non-Ricardian Households and Fiscal Policy in an Estimated DSGE Model of the Euro Area,” *Computing in Economics and Finance* 2005 102, Society for Computational Economics November 2005.
- House, Christopher L. and Matthew D. Shapiro**, “Temporary Investment Tax Incentives: Theory with Evidence from Bonus Depreciation,” *American Economic Review*, 2008, 98 (3), 737–768.
- Jones, John Bailey**, “Has fiscal policy helped stabilize the postwar U.S. economy?,” *Journal of Monetary Economics*, May 2002, 49 (4), 709–746.
- Klein, Paul**, “Using the Generalized Schur Form to Solve a Multivariate Linear Rational Expectations Model,” *Journal of Economic Dynamics and Control*, 2000, 24 (10), 1405–1423.
- Kumhof, Michael, Gunter Coenen, Dirk Muir, Charles Freedman, Susanna Mursula, Christopher J. Erceg, Davide Furceri, Rene Lalonde, Jesper Linde, Annabelle Mourougane, John Roberts, Douglas Laxton, Calos de Resende, Werner Roeger, Stephen Snudden, Mathias Trabandt, and Jan in ’t Veld**, “Effects of Fiscal Stimulus in Structural Models,” IMF Working Papers 10/73, International Monetary Fund March 2010.
- Leeper, Eric M., Michael Plante, and Nora Traum**, “Dynamics of fiscal financing in the United States,” *Journal of Econometrics*, June 2010, 156 (2), 304–321.
- Mendoza, Enrique G., Assaf Razin, and Linda L. Tesar**, “Effective tax rates in macroeconomics: Cross-country estimates of tax rates on factor incomes and consumption,” *Journal of Monetary Economics*, 1994, 34 (3), 297–323.
- Ravn, Morten, Stephanie Schmitt-Grohé, and Martín Uribe**, “Deep Habits,” *The Review of Economic Studies*, 2006, 73 (1), 195–218.
- Smets, Frank and Raf Wouters**, “An Estimated Dynamic Stochastic General Equilibrium Model of the Euro Area,” *Journal of the European Economic Association*, 09 2003, 1 (5), 1123–1175.
- Traum, Nora and Shu-Chun Yang**, “When Does Government Debt Crowd Out Investment,” *CAEPR Working Paper*, May 2010, (#006-2010).
- Zubairy, Sarah**, “On Fiscal Multipliers: Estimates from a Medium Scale DSGE Model,” Working Papers 10-30, Bank of Canada 2010.

Tables

Table 1: Calibrated Parameters

Parameter	Description	Value
<i>Preferences and technology</i>		
β	Discount factor	$1/(1.03)^{1/4}$
δ	Quarterly economic depreciation rate	0.025
θ	Capital income share	0.30
η	Elasticity of substitution, goods varieties	5.3
η_w	Elasticity of substitution, labor varieties	21.0
<i>Steady-state targets</i>		
$\bar{\pi}$	Steady-state gross inflation	1.0
$\bar{h}$	Steady-state hours worked	0.50
$\bar{g}/\bar{y}$	Government spending share of output	0.21
$\bar{b}/\bar{y}$	Debt-to-output ratio	0.32
$\bar{\text{div}}/\bar{y}$	Dividend share of output	0.05
<i>Steady-state tax rates</i>		
$\bar{\tau}^k$	Capital income tax rate	0.42
$\bar{\tau}^w$	Labor income tax rate	0.22
$\bar{\tau}^c$	Consumption tax rate	0.06
$\bar{\tau}^i$	Interest income tax rate	0.25
$\bar{\tau}^d$	Dividend tax rate	0.30
$\bar{\tau}^{ic}$	Investment tax credit rate	0.02
$\bar{e}^\tau$	Immediate expensing rate	0.06
δ^τ	Tax depreciation rate	0.025

Table 2: Estimated Parameters: Prior Distributions

Parameter	Description	Prior	Mean	Std
<i>Deep habits</i>				
b_c	Private consumption habit depth	Beta	0.70	0.10
θ^c	Private habit stock persistence	Beta	0.80	0.10
b_g	Government consumption habit depth	Beta	0.70	0.10
θ^g	Government habit stock persistence	Beta	0.80	0.10
<i>Structural parameters</i>				
γ	Risk aversion / inverse EIS	Normal	2.00	1.00
σ_u	Utilization cost curvature	Normal	2.50	0.50
κ	Investment adjustment cost	Normal	2.00	0.50
α_P	Rotemberg price adjustment cost	Normal	17.0	5.00
α_W	Rotemberg wage adjustment cost	Normal	100.0	30.0
<i>Monetary policy (Taylor rule)</i>				
α_R	Interest rate smoothing	Beta	0.80	0.20
α_π	Inflation response	Normal	1.60	0.20
α_Y	Output response	Normal	0.10	0.05
<i>Exogenous process persistence</i>				
ρ_z	Technology shock	Beta	0.70	0.20
ρ_d	Preference shock	Beta	0.70	0.20
ρ_μ	Investment efficiency shock	Beta	0.70	0.20
<i>Fiscal rule AR coefficients</i>				
ρ_k	Capital tax rule persistence	Beta	0.70	0.20
ρ_w	Labor tax rule persistence	Beta	0.70	0.20
ρ_g	Government spending rule persistence	Beta	0.80	0.20
ρ_{tr}	Transfer rule persistence	Beta	0.70	0.20
ρ_c	Consumption tax rule persistence	Beta	0.70	0.20
ρ_{ic}	ITC rule persistence	Beta	0.70	0.20
ρ_{ti}	Interest income tax rule persistence	Beta	0.70	0.20
ρ_{td}	Dividend tax rule persistence	Beta	0.70	0.20
$\rho_{e\tau}$	Expensing rule persistence	Beta	0.70	0.20
<i>Fiscal rule debt-feedback coefficients</i>				
$\rho_{k,b}$	Capital tax / debt	Gamma	0.50	0.25
$\rho_{w,b}$	Labor tax / debt	Gamma	0.50	0.25
$\rho_{g,b}$	Government spending / debt	Gamma	0.50	0.25
$\rho_{tr,b}$	Transfers / debt	Gamma	0.50	0.25
$\rho_{c,b}$	Consumption tax / debt	Gamma	0.50	0.25
$\rho_{ti,b}$	Interest income tax / debt	Gamma	0.50	0.25
$\rho_{td,b}$	Dividend tax / debt	Gamma	0.50	0.25
$\rho_{ic,b}$	ITC / debt	Normal	0.00	0.25
$\rho_{e\tau,b}$	Expensing / debt	Normal	0.00	0.25
<i>Fiscal rule output-feedback coefficients</i>				
$\rho_{k,y}$	Capital tax / output	Gamma	0.15	0.10
$\rho_{w,y}$	Labor tax / output	Gamma	0.15	0.10
$\rho_{g,y}$	Government spending / output	Normal	-0.05	0.05
$\rho_{tr,y}$	Transfers / output	Normal	-0.10	0.05
$\rho_{c,y}$	Consumption tax / output	Normal	0.00	0.10
$\rho_{ic,y}$	ITC / output	Normal	0.00	0.10
$\rho_{ti,y}$	Interest income tax / output	Normal	0.00	0.10
$\rho_{td,y}$	Dividend tax / output	Normal	0.00	0.10
$\rho_{e\tau,y}$	Expensing / output	Normal	0.00	0.10
<i>Cross-shock correlations</i>				
ϕ	ε_t^k entering labor tax rule	Normal	0.25	0.10
ϕ_{wd}	ε_t^w entering dividend tax rule	Normal	0.25	0.10
<i>Shock standard deviations (all inv-Gamma, mean 0.05, std 0.10)</i>				
$\sigma_{\varepsilon^k}, \sigma_{\varepsilon^w}, \sigma_{\varepsilon^g}, \sigma_{\varepsilon^{tr}}, \sigma_{\varepsilon^z}, \sigma_{\varepsilon^d}, \sigma_{\varepsilon^\mu}, \sigma_{\varepsilon^m}$	Zubairy shocks	Inv-Gamma	0.05	0.10
$\sigma_{\varepsilon^c}, \sigma_{\varepsilon^{ic}}, \sigma_{\varepsilon^{ti}}, \sigma_{\varepsilon^{td}}, \sigma_{\varepsilon^{e\tau}}$	Extension shocks	Inv-Gamma	0.05	0.10
σ_{ε^b}	Debt residual	Inv-Gamma	0.05	0.10

Note: Parameters below the horizontal rule are new to this paper relative to Zubairy (2010). Gamma priors for debt-feedback coefficients of tax rules enforce positive values (taxes rise with debt). Normal priors for debt-feedback coefficients of the ITC and expensing rules are centered at zero because these instruments may either rise or fall in response to debt. Gamma priors for output-feedback coefficients enforce the sign implied by automatic stabilizers for tax rates; Normal priors for spending rules allow the sign to be determined by the data.

Table 3: Fiscal Multipliers, Total-Revenue Normalization

Instrument	$H = 1$	$H = 4$	$H = 12$	$H = 20$
<i>Panel A. Zubairy-original instruments</i>				
Government spending	1.09 [1.04, 1.14]	1.00 [0.93, 1.07]	0.59 [0.52, 0.65]	0.35 [0.26, 0.42]
Labor tax	0.22 [0.19, 0.26]	0.47 [0.41, 0.52]	0.79 [0.65, 0.91]	0.70 [0.53, 0.87]
Capital tax	0.90 [0.81, 0.97]	1.53 [1.30, 1.66]	3.08 [2.33, 3.51]	2.89 [2.18, 3.28]
<i>Panel B. Extension instruments</i>				
Consumption tax	0.28 [0.26, 0.30]	0.41 [0.37, 0.44]	0.26 [0.05, 0.43]	−0.05 [−0.31, 0.19]
Investment tax credit	0.87 [0.71, 1.09]	2.08 [1.57, 2.70]	3.71 [2.56, 4.50]	3.43 [2.54, 3.99]
Expensing rate	0.28 [0.24, 0.30]	0.52 [0.44, 0.57]	0.65 [0.46, 0.87]	0.61 [0.31, 0.90]
Dividend tax	0.27 [0.19, 0.34]	0.30 [0.20, 0.40]	−0.01 [−0.11, 0.09]	−0.28 [−0.42, −0.15]
Interest income tax	−1.87 [−1.98, −1.73]	−2.21 [−2.30, −2.07]	−2.93 [−3.49, −2.48]	−3.88 [−5.27, −3.03]
Transfers	0.08 [0.07, 0.10]	0.14 [0.10, 0.18]	−0.20 [−0.49, 0.06]	−0.44 [−0.72, −0.18]
<i>Panel C. Zubairy (2010), Table 2</i>				
Government spending	1.07 [1.01, 1.13]	1.06 [0.96, 1.17]	0.90 [0.73, 1.07]	0.72 [0.49, 0.93]
Labor tax	0.13 [0.09, 0.18]	0.32 [0.22, 0.45]	0.68 [0.41, 1.09]	0.85 [0.42, 1.58]
Capital tax	0.34 [0.30, 0.37]	0.43 [0.34, 0.51]	0.52 [0.30, 0.73]	0.46 [0.15, 0.81]

Note: Present-value output multipliers computed under the total-revenue normalization (equation 2.2). Posterior median with 5–95 percentile intervals in brackets, from 1,000 draws sampled from the posterior of the model estimated over 1961Q1–2013Q4. The multiplier is expressed in dollars of cumulative output per dollar of cumulative revenue change, with sign convention such that a positive value indicates output moves in the expansionary direction of the policy.

Table 4: Fiscal Multipliers, Own-Revenue Normalization

Instrument	$H = 1$	$H = 4$	$H = 12$	$H = 20$
<i>Panel A. Zubairy-original instruments</i>				
Government spending	0.29 [0.28, 0.31]	1.00 [0.93, 1.07]	1.51 [1.28, 1.68]	1.34 [0.92, 1.65]
Labor tax	0.07 [0.06, 0.08]	0.43 [0.38, 0.48]	0.99 [0.75, 1.19]	0.99 [0.69, 1.29]
Capital tax	0.38 [0.35, 0.40]	1.83 [1.69, 1.96]	4.70 [4.09, 5.20]	5.82 [4.80, 6.68]
<i>Panel B. Extension instruments</i>				
Consumption tax	0.09 [0.09, 0.10]	0.34 [0.31, 0.37]	0.20 [0.04, 0.37]	-0.04 [-0.22, 0.17]
Investment tax credit	0.20 [0.16, 0.25]	1.27 [1.05, 1.50]	3.20 [2.95, 3.40]	4.11 [3.66, 4.57]
Expensing rate	0.08 [0.07, 0.09]	0.45 [0.39, 0.49]	0.77 [0.59, 0.93]	0.74 [0.43, 1.03]
Dividend tax	0.06 [0.05, 0.08]	0.25 [0.18, 0.32]	-0.01 [-0.20, 0.15]	-0.64 [-1.01, -0.35]
Interest income tax	-1.14 [-1.22, -1.06]	-5.13 [-5.65, -4.61]	-9.03 [-10.49, -7.34]	-9.82 [-11.93, -7.67]
Transfers	0.03 [0.03, 0.04]	0.14 [0.10, 0.18]	-0.14 [-0.35, 0.04]	-0.39 [-0.63, -0.15]

Note: Present-value output multipliers computed under the own-revenue normalization (equation 2.3). Denominator is the year-1 discounted cumulative change in the revenue collected by instrument j (for the ITC, the outlay $\tau_t^{ic} i_t$; for the expensing rate, the change in capital tax revenue). Otherwise as in Table 3.

Figures

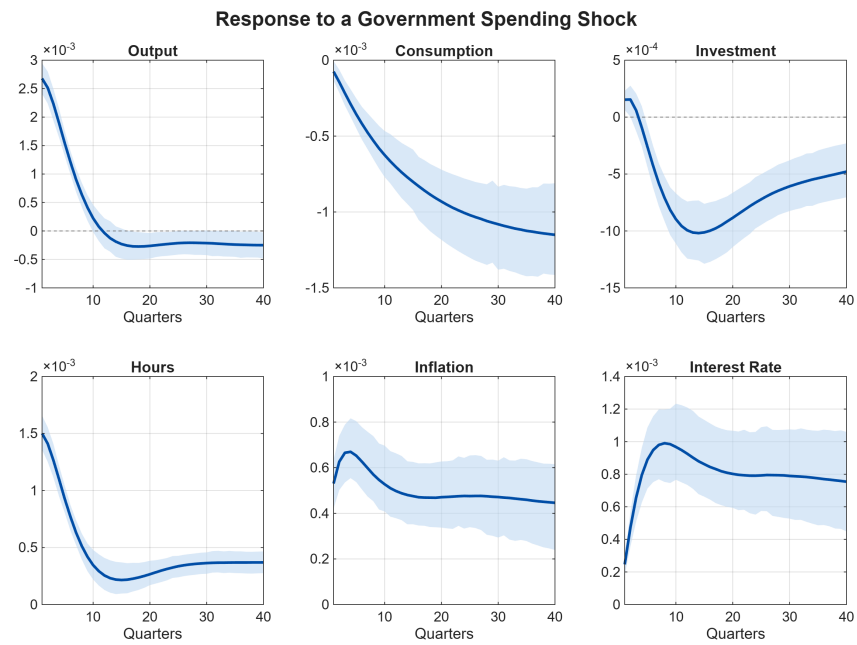


Figure 1: Impulse Responses to a Government Spending Shock



Figure 2: Impulse Responses to a Labor Income Tax Shock

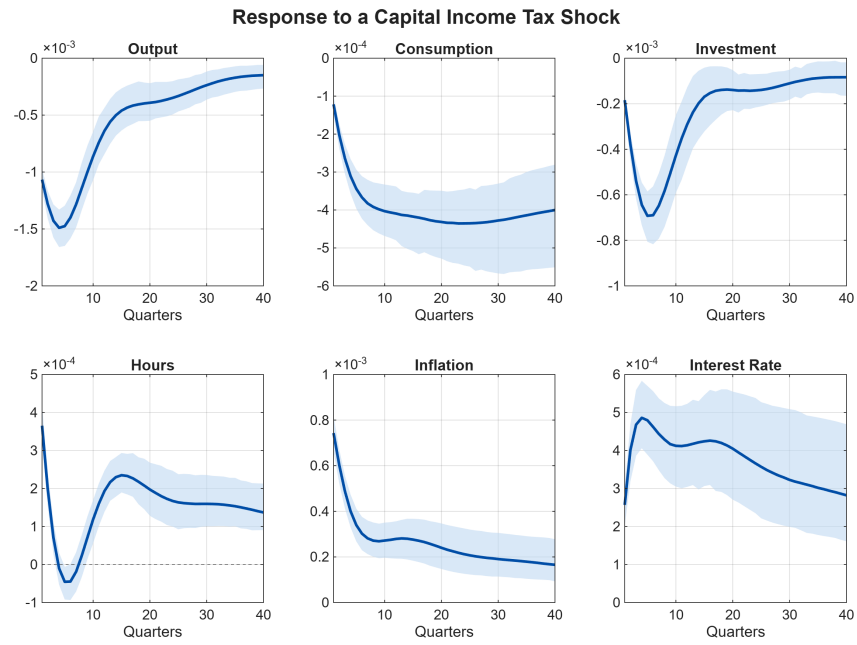


Figure 3: Impulse Responses to a Capital Income Tax Shock

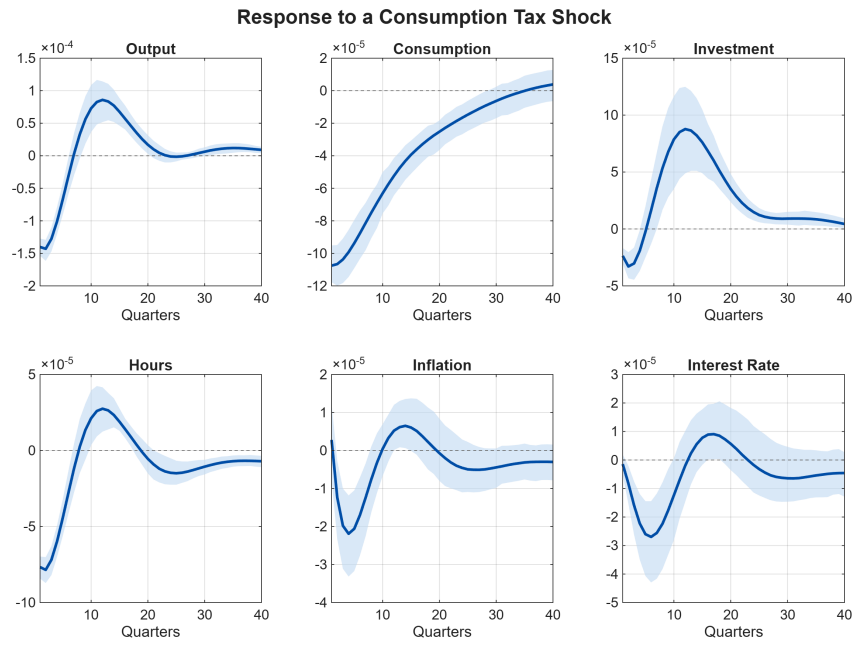


Figure 4: Impulse Responses to a Consumption Tax Shock

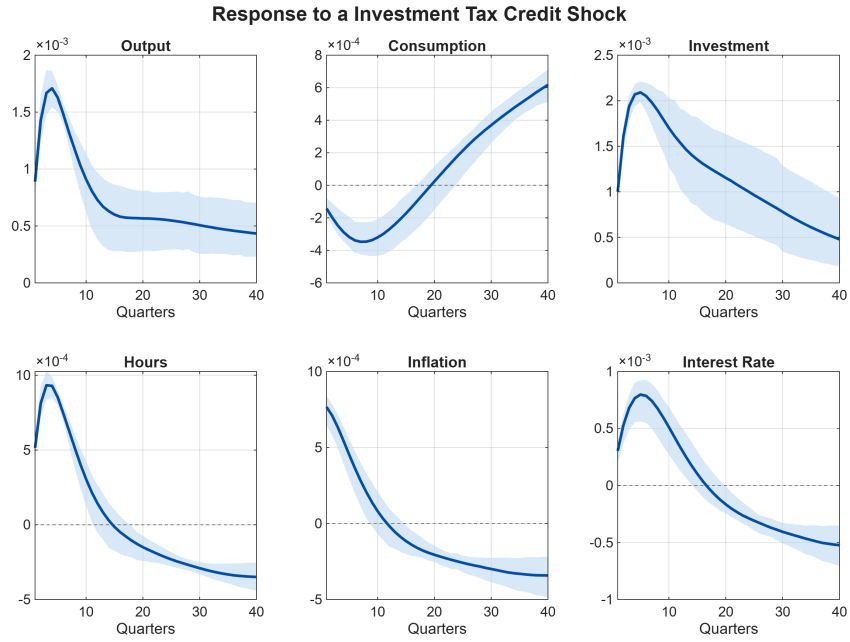


Figure 5: Impulse Responses to an Investment Tax Credit Shock

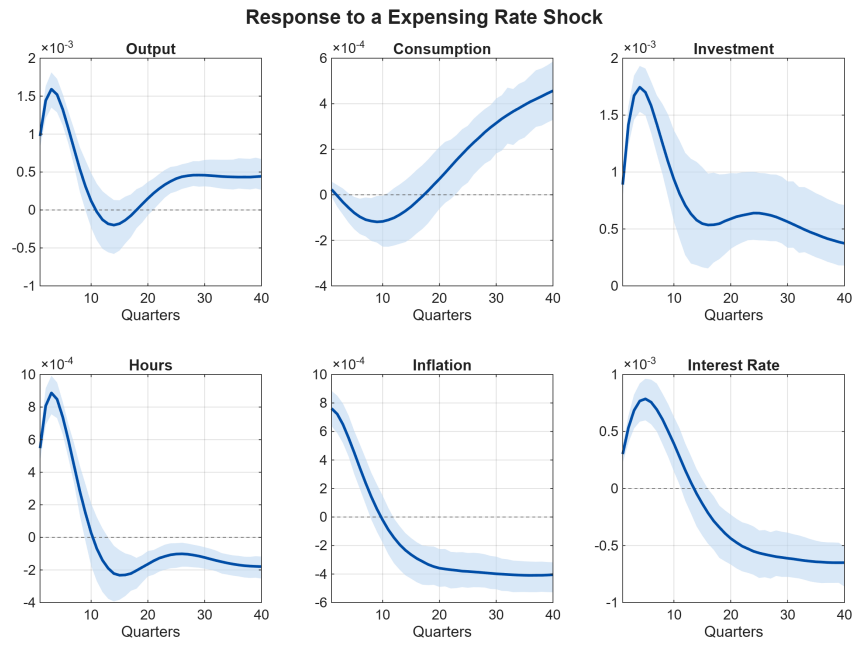


Figure 6: Impulse Responses to an Expensing Rate Shock

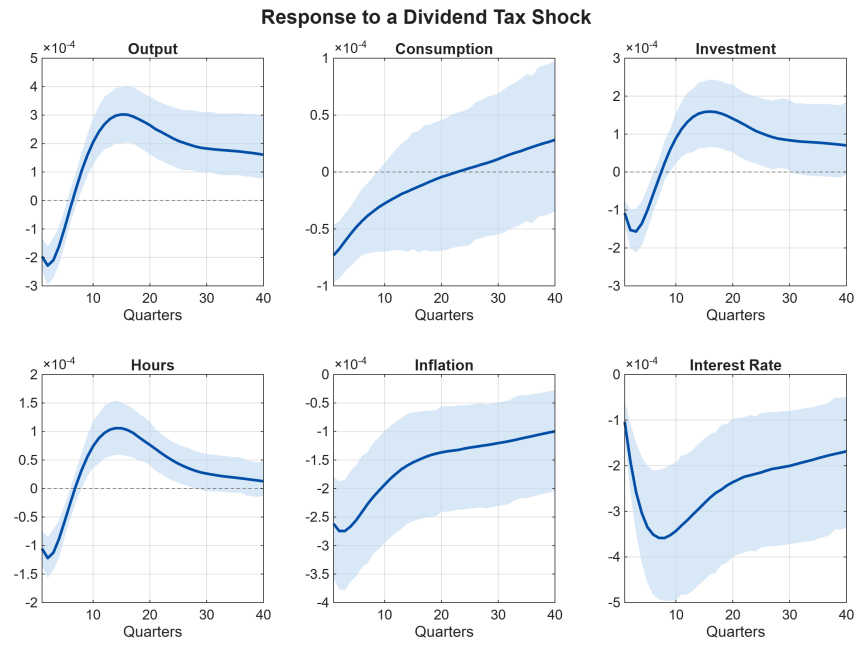


Figure 7: Impulse Responses to a Dividend Tax Shock

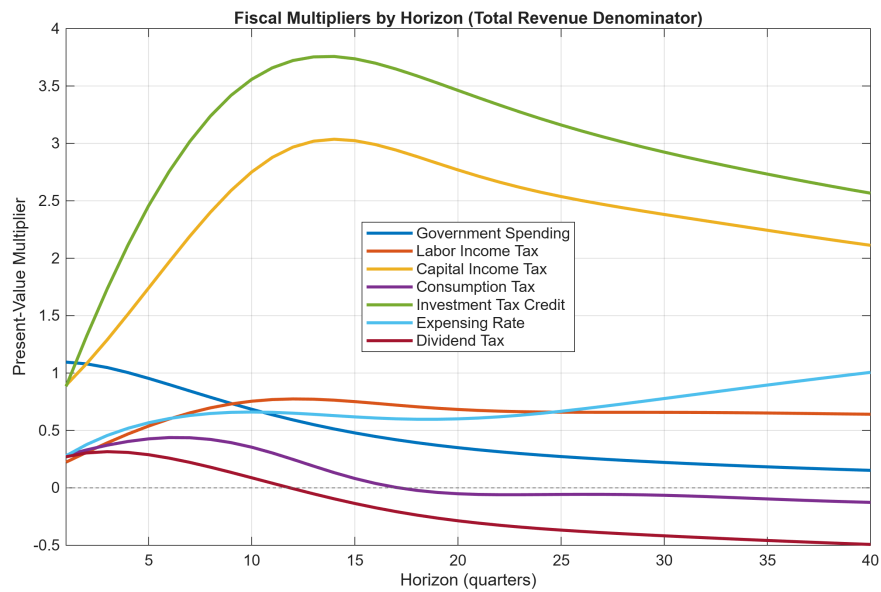


Figure 8: Present-Value Multipliers by Horizon, Total-Revenue Normalization

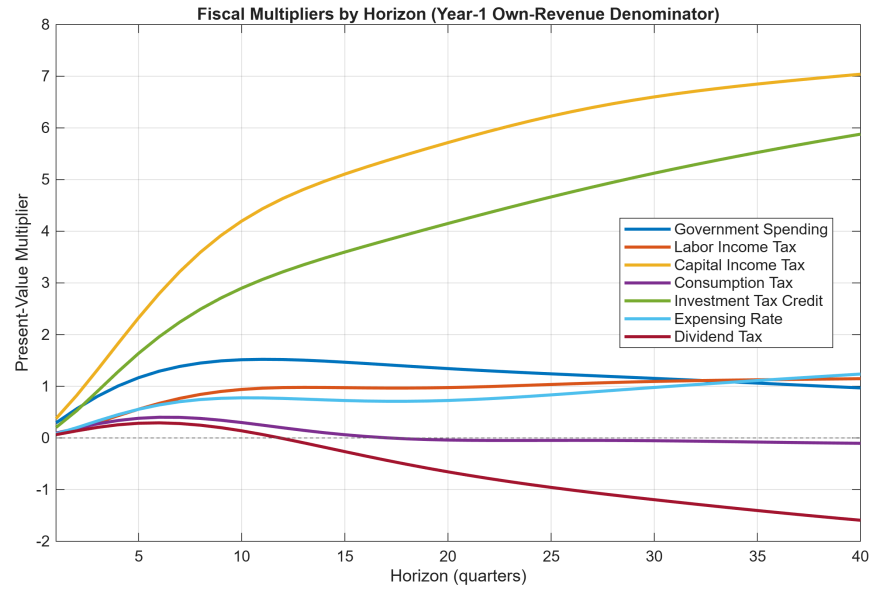


Figure 9: Present-Value Multipliers by Horizon, Own-Revenue Normalization

Appendix

A-1 Household First-Order Conditions

This appendix derives the household first-order conditions of the model presented in Section 1. Zubairy's original derivations, which apply unchanged to the parts of the model that are not modified by our tax extensions, are reproduced in Zubairy (2010). We restrict attention here to the conditions that are either new or modified relative to the baseline.

A-1.1 Utility and Constraints

The representative household maximizes:

$$E_0 \sum_{t=0}^{\infty} \beta^t d_t U(x_t^c, h_t) \quad \text{with} \quad U(x_t^c, h_t) = \frac{1}{1-\gamma} [(x_t^c)^a (1-h_t)^{1-a}]^{1-\gamma} \quad (\text{A.1.1})$$

where $x_t^c = c_t - b^c s_{t-1}^C$ is habit-adjusted consumption, $s_t^C = \theta^c s_{t-1}^C + (1-\theta^c)c_t$ is the private habit stock, a is pinned down in steady state, γ is risk aversion, and $b^c \in (0,1)$ is the deep-habit parameter.

The household faces the flow budget constraint:

$$(1 + \tau_t^c)c_t + \frac{b_t}{\pi_t} + (1 - \tau_t^{ic} - \tau_t^k e_{\tau,t})i_t = (1 - \tau_t^w)\frac{w_t}{\tilde{w}_t}h_t + (1 - \tau_t^k)r_t^k u_t k_{t-1} + \tau_t^k \delta_{\tau} k_{t-1}^{\tau} \\ + \frac{[1 + (R_{t-1} - 1)(1 - \tau_t^i)] b_{t-1}}{\pi_t} + (1 - \tau_t^d)div_t + tr_t \quad (\text{A.1.2})$$

the capital accumulation law:

$$k_t = (1 - \delta)k_{t-1} + i_t \left[1 - \frac{\kappa}{2} \left(\frac{\mu_t i_t}{i_{t-1}} - 1 \right)^2 \right] \quad (\text{A.1.3})$$

and the tax-basis accumulation law (new):

$$k_t^{\tau} = (1 - \delta_{\tau})k_{t-1}^{\tau} + i_t(1 - e_{\tau,t}) \quad (\text{A.1.4})$$

The Lagrange multipliers on (A.1.2), (A.1.3), and (A.1.4) are λ_t , $\lambda_t q_t$, and $\lambda_t v_{\tau,t}$ respectively. In Zubairy's baseline the tax basis of capital is not separated from the physical capital stock; here we introduce it because immediate expensing (e_{τ}) drives a wedge between the two.

A-1.2 Marginal Utility of Consumption

The FOC with respect to c_t gives $\partial U / \partial c_t = (1 + \tau_t^c)\lambda_t$. Substituting the marginal utility from (A.1.1):

$$\lambda_t = d_t a \frac{[(x_t^c)^a (1-h_t)^{1-a}]^{1-\gamma}}{x_t^c (1 + \tau_t^c)} \quad (\text{A.1.5})$$

Relative to Zubairy (2010), the $(1 + \tau_t^c)$ factor in the denominator is new. Because λ_t enters every forward-looking equation of the household problem, the consumption tax propagates into the bond Euler, capital Euler, investment FOC, wage Phillips curve, and pricing block without changing the functional form of those equations.

A-1.3 Labor–Leisure Condition

The FOC with respect to h_t combined with (A.1.5) gives the intratemporal condition:

$$d_t(1-a) \frac{[(x_t^c)^a(1-h_t)^{1-a}]^{1-\gamma}}{1-h_t} = \lambda_t \frac{w_t(1-\tau_t^w)}{\tilde{w}_t} \quad (\text{A.1.6})$$

This is functionally identical to Zubairy’s condition after multiplication by $(1+\tau_t^c)$, which reappears through the modified λ_t .

A-1.4 Bond Euler Equation

Nominal bonds pay R_t in period $t+1$ per dollar invested, but only the interest earned, $R_t - 1$, is taxable. Combining the FOC with respect to b_t and applying (A.1.5):

$$\lambda_t = \beta [1 + (R_t - 1)(1 - \tau_{i,t+1})] \frac{\lambda_{t+1}}{\pi_{t+1}} \quad (\text{A.1.7})$$

Under $\tau_i = 0$ this collapses to Zubairy’s Euler equation.

A-1.5 Utilization and Capital Euler

The FOC with respect to utilization u_t delivers:

$$(1 - \tau_t^k) r_t^k = \gamma_1 u_t^{\sigma_u} \quad (\text{A.1.8})$$

The FOC with respect to k_t (using the capital constraint) gives:

$$\lambda_t q_t = \beta E_t \lambda_{t+1} \left[(1 - \tau_{k,t+1}) r_{t+1}^k u_{t+1} - \frac{\gamma_1}{1+\sigma_u} (u_{t+1}^{1+\sigma_u} - 1) + q_{t+1}(1 - \delta) \right] \quad (\text{A.1.9})$$

The depreciation deduction $\delta q_{t+1} \tau_{t+1}^k u_{t+1}$ present in Zubairy (2010), which credits the capital-tax shield directly in the Euler, is absent here. Instead the shield is carried entirely through the shadow value $v_{\tau,t}$ derived below, avoiding double-counting.

A-1.6 Shadow Value of the Tax Basis

The FOC with respect to k_t^τ delivers a forward-looking recursion. Because an extra unit of tax basis today reduces next-period taxes by $\tau_{k,t+1} \delta_\tau$ and leaves $(1 - \delta_\tau)$ units of remaining basis, we obtain:

$$\lambda_t v_{\tau,t} = \beta E_t [\lambda_{t+1} ((1 - \delta_\tau) v_{\tau,t+1} + \tau_{k,t+1} \delta_\tau)] \quad (\text{A.1.10})$$

This equation is new relative to Zubairy (2010) and is what makes it possible to separate immediate expensing (through e_τ) from the ordinary depreciation schedule.

A-1.7 Investment First-Order Condition

The FOC with respect to i_t is the most extensively modified condition. The out-of-pocket cost of a unit of investment falls by τ_t^{ic} (the ITC) and by $\tau_t^k e_{\tau,t}$ (the immediate expensing

wedge). The non-expensed portion $(1 - e_{\tau,t})$ generates future depreciation deductions valued at $v_{\tau,t}$:

$$\begin{aligned} \lambda_t(1 - \tau_t^{ic} - \tau_t^k e_{\tau,t}) &= \lambda_t q_t \left[1 - \frac{\kappa}{2} \left(\frac{\mu_t i_t}{i_{t-1}} - 1 \right)^2 - \kappa \frac{\mu_t i_t}{i_{t-1}} \left(\frac{\mu_t i_t}{i_{t-1}} - 1 \right) \right] \\ &\quad + \beta E_t \left[\lambda_{t+1} q_{t+1} \kappa \frac{\mu_{t+1} i_{t+1}^2}{i_t^2} \left(\frac{\mu_{t+1} i_{t+1}}{i_t} - 1 \right) \right] \\ &\quad + \lambda_t v_{\tau,t} (1 - e_{\tau,t}) \end{aligned} \tag{A.1.11}$$

Under $\tau^{ic} = e_{\tau} = 0$ the LHS and last term collapse to Zubairy's original expression.

A-1.8 Dividend Tax

The dividend tax τ_t^d does not appear in any household first-order condition. Dividends are a residual object taken as given by the household:

$$div_t = y_t - w_t h_t - r_t^k u_t k_{t-1} \tag{A.1.12}$$

The dividend tax only affects the household's after-tax income (equation [A.1.2](#)) and enters the government budget constraint through tax revenue.

A-2 Steady-State Derivations

We compute the deterministic steady state analytically by setting all shocks to their unconditional means ($z = d = \mu = 1$; all fiscal innovations equal to zero) and imposing stationarity. Below, a bar denotes a steady-state value and we drop time subscripts.

A-2.1 Shadow Value of the Tax Basis in Steady State

Setting the recursion (A.1.10) to a fixed point and using $\bar{\lambda}_t = \bar{\lambda}_{t+1} = \bar{\lambda}$:

$$\bar{v}_\tau = \frac{\beta \bar{\tau}^k \delta_\tau}{1 - \beta(1 - \delta_\tau)} \quad (\text{A.2.1})$$

which is the present value of the geometric stream of depreciation deductions on one unit of tax basis. This expression is valid for any positive $\bar{\tau}^k$ and δ_τ , and does not depend on the immediate expensing rate.

A-2.2 Tobin's q in Steady State

Setting the investment FOC (A.1.11) to a fixed point (all adjustment cost terms vanish because $\bar{\mu} = 1$ and $\bar{i}_t = \bar{i}_{t-1}$):

$$\bar{q} = 1 - \bar{\tau}^{ic} - \bar{\tau}^k \bar{e}^\tau - \bar{v}_\tau(1 - \bar{e}^\tau) \quad (\text{A.2.2})$$

Under $\bar{\tau}^{ic} = 0$ and $\bar{e}^\tau = 0$, this collapses to $\bar{q} = 1 - \bar{v}_\tau$, matching the geometric-stream expression in Zubairy's baseline.

A-2.3 Real Return on Capital, Real Interest Rate, and Real Wage

The steady-state capital Euler (A.1.9) combined with $\bar{u} = 1$ implies:

$$\bar{r}^k = \frac{\bar{q}}{\beta(1 - \tau^k)} - \frac{q(1 - \delta)}{1 - \tau^k} \quad (\text{A.2.3})$$

The steady-state bond Euler (A.1.7) with $\bar{\pi} = 1$ delivers:

$$\bar{R} = 1 + \frac{1/\beta - 1}{1 - \bar{\tau}^i} \quad (\text{A.2.4})$$

The real wage follows from the firm's optimization conditions and, given the Cobb–Douglas production function, from the capital–labor ratio and marginal cost:

$$\bar{w} = (1 - \theta) \bar{m}\bar{c} \left(\frac{\bar{r}^k}{\theta \bar{m}\bar{c}} \right)^{-\theta/(1-\theta)} \quad (\text{A.2.5})$$

where $\bar{m}\bar{c} = (\eta - 1)/\eta$ is pinned down by the desired steady-state markup.

A-2.4 Aggregate Ratios

Given $\bar{h}$, the steady-state capital stock and output follow:

$$\frac{\bar{k}}{\bar{h}} = \left(\frac{\theta \bar{m} \bar{c}}{\bar{r}^k} \right)^{1/(1-\theta)}, \quad \bar{y} = \bar{k}^\theta \bar{h}^{1-\theta} - \psi \quad (\text{A.2.6})$$

The fixed operating cost ψ is set residually to hit the target dividend share $\overline{div}/\bar{y}$; steady-state investment is $\bar{i} = \delta \bar{k}$; steady-state consumption is $\bar{c} = \bar{y} - \bar{i} - \bar{g}$, using the resource constraint with steady-state utilization cost equal to zero; and steady-state government purchases and debt are determined by the calibrated ratios $\bar{g}/\bar{y}$ and $\bar{b}/\bar{y}$.

A-2.5 Marginal Utility in Steady State

The steady state of the marginal utility equation (A.1.5) pins down the consumption share parameter a jointly with $\bar{\lambda}$. Given $\bar{h}$, $\bar{x}^c = (1 - b^c)\bar{c}$, and the labor-leisure condition (A.1.6), we solve:

$$a = \frac{\bar{x}^c(1 + \bar{\tau}^c)}{\bar{x}^c(1 + \bar{\tau}^c) + (\bar{w}/\bar{\tilde{w}})(1 - \bar{h})}, \quad \bar{\lambda} = a \frac{[(\bar{x}^c)^a(1 - \bar{h})^{1-a}]^{1-\gamma}}{\bar{x}^c(1 + \bar{\tau}^c)} \quad (\text{A.2.7})$$

Both quantities depend on estimated parameters and are recomputed at each draw during Bayesian inference.

A-2.6 Government Budget in Steady State

Setting the government budget constraint (1.31) to a stationary value $\bar{b}$ and using $\bar{\pi} = 1$:

$$\bar{b}(1 - \bar{R}) + \overline{taxrev} = \bar{g} + \bar{tr} \quad (\text{A.2.8})$$

Given the target debt-to-output ratio and the calibrated tax rates, this determines the steady-state transfer $\bar{tr}$ residually. The debt-residual shock ε_t^b has mean zero and therefore does not appear in the steady state.